

GRAVEATLAS

Phase 13 — Autonomous Operations, Intelligence & Self-Healing Platform

GRAVEATLAS – PHASE 13
AUTONOMOUS OPERATIONS, INTELLIGENCE & SELF-HEALING PLATFORM

PHASE 1 THROUGH PHASE 12 MUST ALREADY BE COMPLETE.

Continue from the existing GraveAtlas project.

=====
CORE RULES
=====

- Do NOT rebuild Phases 1-12.
- Do NOT start Phase 14.
- Reuse existing architecture, security, provenance, moderation, audit, preservation, APIs, datasets and release systems.
- No paid services or external integrations without explicit approval.
- Prefer existing/free/local processing where practical.
- Never expose secrets, credentials, tokens or private data.
- Never fabricate historical facts, records, users, analytics, coverage, performance or test results.
- Never silently alter authoritative historical records.
- AI recommendations are not authoritative historical facts.
- Do not perform destructive or mass operations without authorization.
- Complete Phase 13 only, then stop.

=====
PART 1 – PHASE 12 BASELINE
=====

Read the actual Phase 12 implementation and final report.

Identify:

- completed capabilities
- remaining blockers
- actual limitations
- preservation health
- interoperability status
- API maturity
- governance maturity
- operational risks

Use evidence only.

=====
PART 2 – AUTONOMOUS SYSTEM MONITORING
=====

Implement or improve monitoring for:

- application health
- backend/API health
- database health
- search/index health
- map services
- scheduled jobs
- data pipelines
- backup/preservation systems

Use actual measurements.

=====
PART 3 – HEALTH CHECKS
=====

Create safe health checks for:

- availability
- latency
- errors
- failed jobs
- stale data
- failed indexing
- backup status
- storage capacity

Do not create false-positive dashboards.

=====
PART 4 – SAFE SELF-HEALING
=====

Where technically safe, automate recovery for reversible technical failures.

Examples:

- restart recoverable service
- retry failed non-destructive job
- rebuild a derived search index
- refresh a safe cache
- resume interrupted processing

Every automated action must be:

- bounded
- authorized
- logged
- observable
- reversible where applicable

Never automatically rewrite authoritative historical data.

=====
PART 5 – FAILURE ESCALATION
=====

If automatic recovery fails:

- DETECT
- CLASSIFY
 - ATTEMPT SAFE RECOVERY
 - VERIFY
 - ESCALATE
 - LOG

Do not endlessly retry failures.

=====
PART 6 – DATA QUALITY ENGINE
=====

Detect potential:

- duplicate records
- duplicate cemeteries
- suspicious coordinates
- inconsistent dates
- missing sources
- broken provenance
- anomalous submissions
- schema violations

Detection must create reviewable findings rather than silently changing data.

=====
PART 7 – AI DATA REVIEW
=====

AI may assist in identifying:

- possible duplicates
- OCR errors
- suspicious values
- possible relationships
- source inconsistencies

AI must provide:

- evidence
- explanation
- uncertainty/confidence
- recommendation

Human review is required before authoritative historical changes.

=====
PART 8 – PROVENANCE VERIFICATION
=====

Automatically check where practical:

- missing source references
- broken source links
- orphaned provenance
- inconsistent source relationships
- unsupported claims
- invalid provenance chains

Do not invent replacement sources.

=====
PART 9 – AUTOMATED PRESERVATION
=====

Where existing infrastructure supports it, automate:

- dataset snapshots

- integrity checks
- backup verification
- archive health checks
- restoration readiness checks

Record:

- timestamp
- version
- result
- errors
- verification status

=====

PART 10 – INTELLIGENT SEARCH

=====

Evaluate safe intelligent search capabilities:

- natural-language queries
- semantic discovery
- query expansion
- evidence-aware retrieval
- related-record discovery

Search results must remain traceable to actual GraveAtlas data.

AI must not invent search results.

=====

PART 11 – RESEARCH ASSISTANT

=====

If justified, provide research assistance that can:

- understand research questions
- locate relevant GraveAtlas records
- explain source relationships
- summarize available evidence
- identify evidence gaps

The assistant must clearly distinguish:

CONFIRMED
INFERRED
UNKNOWN

It must never fabricate missing evidence.

=====

PART 12 – MODERATION ASSISTANCE

=====

AI/automation may:

- detect spam
- identify possible duplicates
- flag suspicious submissions
- prioritize review queues

Do not automatically make uncertain historical claims authoritative.

=====

PART 13 – ANOMALY DETECTION

=====

Detect unusual:

- data changes
- contribution patterns
- API usage
- traffic
- search failures
- indexing failures
- backup behavior
- authentication activity

Use thresholds based on actual system behavior.

=====

PART 14 – INCIDENT RESPONSE

=====

Create a workflow:

DETECT
→ CLASSIFY
→ CONTAIN
→ RECOVER
→ VERIFY
→ DOCUMENT
→ LEARN

Preserve incident evidence.

=====

PART 15 – EXPLAINABLE AUTOMATION

=====

Every important automated decision should record:

- action
- reason
- evidence
- confidence/uncertainty
- timestamp
- system/component
- result
- human override where applicable

=====

PART 16 – ZERO-TRUST AUTOMATION

=====

Apply:

- least privilege
- scoped credentials
- isolated jobs
- limited permissions
- audit logging
- secret protection

No autonomous process receives unrestricted access.

=====

PART 17 – COST-AWARE AUTOMATION

=====

Prefer:

- local processing
- existing infrastructure
- cached results
- deterministic processing

Avoid unnecessary cloud/API/AI calls.

Require explicit approval for paid services.

=====

PART 18 – AUTONOMOUS OPERATIONS DASHBOARD

=====

Where justified, provide an internal dashboard showing:

- system health
- data health
- preservation health
- security events
- AI review queue
- automation status
- incidents
- failed jobs
- pending human decisions

Protect internal operational information.

=====

PART 19 – HUMAN OVERRIDE

=====

Authorized humans must be able to:

- stop automation
- disable affected workflows
- reject AI recommendations
- quarantine suspicious data
- review findings
- reverse safe automated actions

Human override actions must be audited.

=====

PART 20 – AUTOMATION SAFETY LIMITS

=====

Every autonomous workflow must define:

- scope
- permissions
- maximum retries
- timeout
- failure behavior
- escalation

- rollback/recovery
- audit logging

Never allow uncontrolled autonomous loops.

=====
PART 21 – SECURITY AT THE AUTOMATION LAYER
=====

Review:

- automation credentials
- job isolation
- privilege escalation
- webhook security
- scheduled jobs
- AI tool access
- API access
- secret handling

=====
PART 22 – AI GOVERNANCE
=====

Track where AI is used:

- model/provider
- model version where available
- prompt/workflow version
- source inputs
- generated output
- review decision

Do not send private or sensitive information to external AI without approval.

=====
PART 23 – FALSE-POSITIVE CONTROL
=====

Measure and review false positives for:

- duplicate detection
- anomaly detection
- spam detection
- OCR review
- provenance warnings

Do not optimize solely for automation volume.

=====
PART 24 – HUMAN WORKLOAD
=====

Measure whether automation actually reduces useful human work.

Track where available:

- review queue size
- average review burden
- false-positive burden
- unresolved findings
- recovery success

Never fabricate measurements.

=====
PART 25 – TESTING
=====

Test autonomous workflows safely.

Include:

- normal conditions
- transient failures
- permanent failures
- partial failures
- duplicate events
- timeout conditions
- permission failures
- recovery failures

Use staging/non-production where possible.

=====
PART 26 – DOCUMENTATION
=====

Create/update:

docs/AUTONOMOUS-OPERATIONS.md

docs/HEALTH-MONITORING.md
docs/SELF-HEALING.md
docs/DATA-QUALITY-ENGINE.md
docs/AI-DATA-REVIEW.md
docs/PROVENANCE-VERIFICATION.md
docs/AUTOMATED-PRESERVATION.md
docs/INTELLIGENT-SEARCH.md
docs/RESEARCH-ASSISTANT.md
docs/MODERATION-AUTOMATION.md
docs/ANOMALY-DETECTION.md
docs/INCIDENT-RESPONSE.md
docs/EXPLAINABLE-AUTOMATION.md
docs/AUTOMATION-SECURITY.md
docs/AI-GOVERNANCE.md
docs/PHASE-13-ROADMAP.md

Document actual implementation only.

=====

PHASE 13 ACCEPTANCE GATE

=====

[] Phase 12 baseline reviewed
[] Autonomous monitoring
[] Health checks
[] Safe self-healing
[] Failure escalation
[] Data-quality engine
[] AI data review
[] Provenance verification
[] Automated preservation
[] Intelligent search evaluation
[] Research assistant
[] Moderation assistance
[] Anomaly detection
[] Incident response
[] Explainable automation
[] Zero-trust automation
[] Cost-aware automation
[] Operations dashboard
[] Human override
[] Automation safety limits
[] Automation-layer security
[] AI governance
[] False-positive control
[] Human workload measurement
[] Testing
[] Documentation

=====

FINAL REPORT

=====

PHASE 13 STATUS: READY / NOT READY

For every implemented capability report:

- what was built
- evidence
- tests performed
- limitations
- security considerations
- privacy considerations
- provenance considerations
- operational impact

Use actual metrics only.

=====

BLOCKERS

=====

List every issue preventing Phase 13 from being READY.

For each:

- severity
- component
- issue
- evidence
- impact
- remediation
- status

Never include secrets.

=====

AUTOMATION SCORECARD

=====

Monitoring:
[actual assessment]

Recovery:
[actual assessment]

Data quality:
[actual assessment]

Provenance:
[actual assessment]

AI safety:
[actual assessment]

Security:
[actual assessment]

Preservation:
[actual assessment]

Human oversight:
[actual assessment]

Operational reliability:
[actual assessment]

Do not invent numeric scores without a defined measurement method.

=====

ROADMAP

=====

NOW:
[items]

NEXT:
[items]

LATER:
[items]

DO NOT BUILD YET:
[items]

=====

STOP CONDITION

=====

Do NOT start Phase 14.

Do NOT add paid services without explicit approval.

Do NOT expose secrets.

Do NOT fabricate historical information.

Do NOT silently modify authoritative records.

Do NOT allow AI-generated historical claims to become authoritative without review.

Do NOT perform destructive bulk operations.

Do NOT perform uncontrolled production load testing.

Do NOT allow autonomous workflows unrestricted permissions.

STOP after the Phase 13 final report.

WAIT FOR EXPLICIT INSTRUCTIONS FOR PHASE 14.